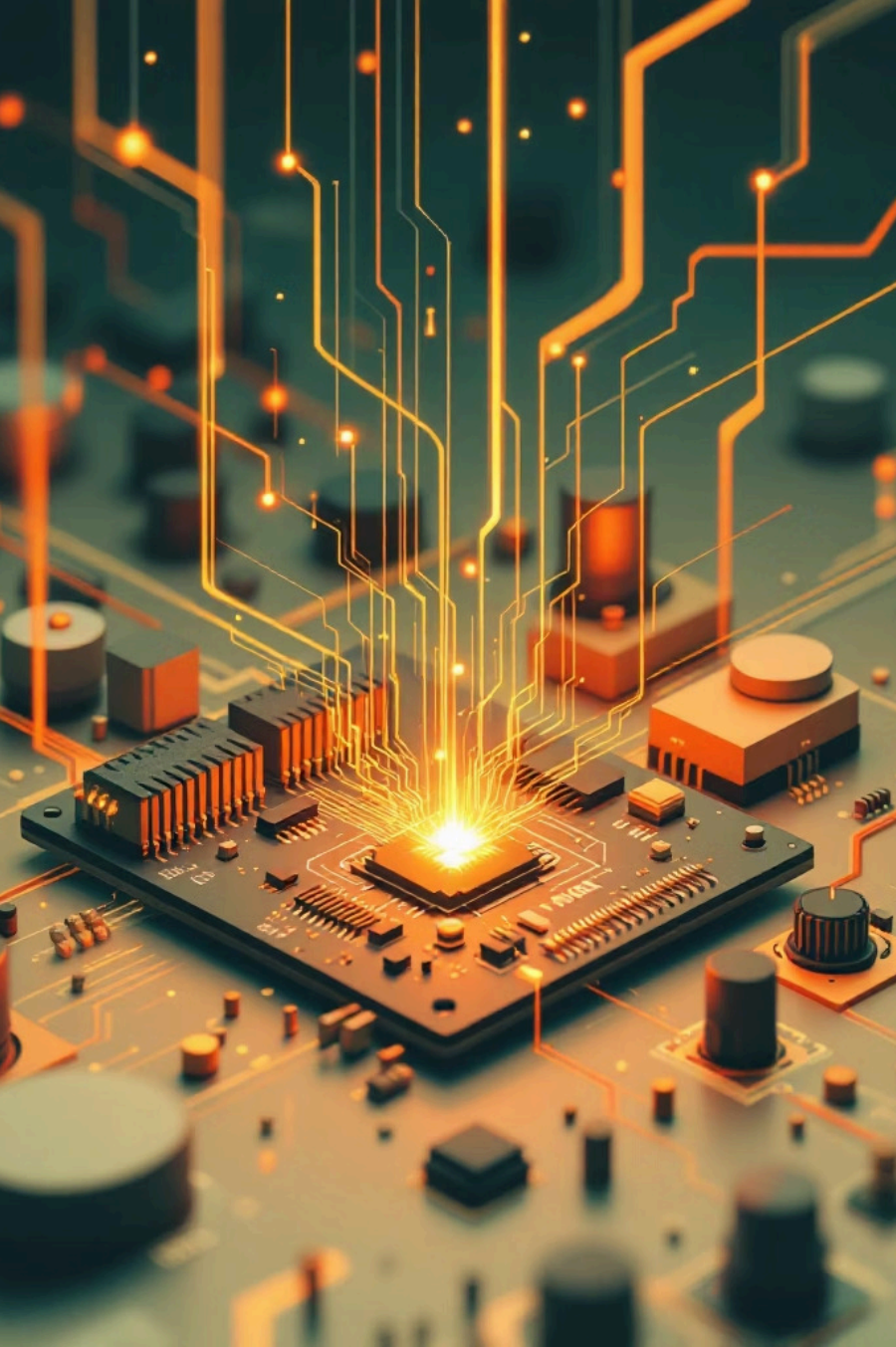




Analysis of 2 kV HBM ESD Event in CMOS

ESD Protection using Double Diode Clamp

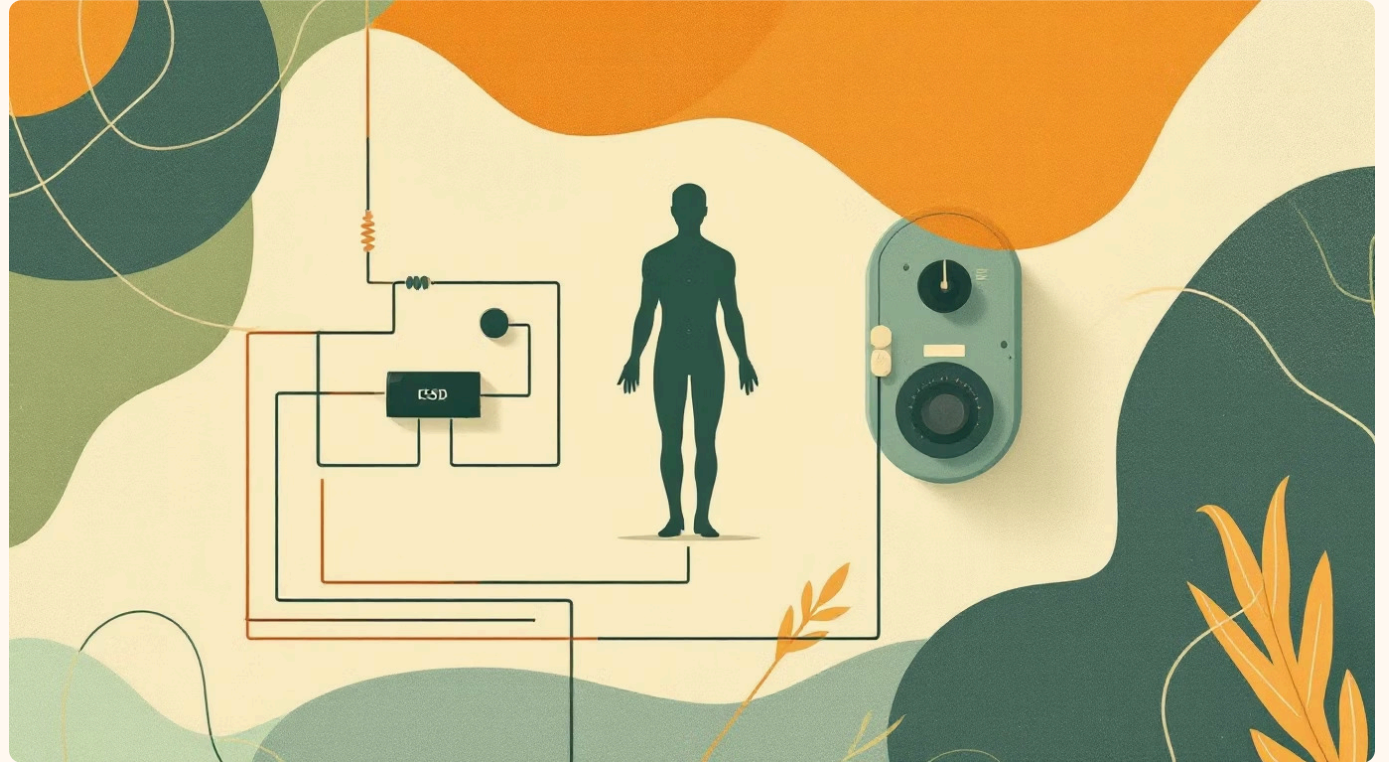
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What is ESD and Why Should We Care?

Electrostatic Discharge (ESD) is a sudden, high-current transfer of electric charge that can permanently damage integrated circuits. Even walking across a carpet can build up enough charge to fry sensitive CMOS transistors.

The Human Body Model (HBM) simulates this by modeling the human body as a simple RC circuit:

- Capacitor: $C = 100 \text{ pF}$
- Resistor: $R = 1.5 \text{ k}\Omega$



CMOS circuits operate at low voltages (typically 1-5 V) and can't handle voltage spikes from ESD events. That's where protection circuits come in.

The Problem: 2 kV HBM Event



Given Voltage

$$V = 2000 \text{ V}$$



Capacitance

$$C = 100 \text{ pF}$$



Resistance

$$R = 1.5 \text{ k}\Omega$$

We need to figure out what happens when this 2 kV discharge hits our CMOS circuit. Specifically, we want to find:

- Current waveform over time
- Peak current magnitude
- Total energy dissipated
- How double diode clamps protect the circuit
- Whether latch-up is a risk

Let's Do the Math

For an RC discharge circuit, the current follows an exponential decay:

$$i(t) = \frac{V_0}{R} e^{-t/RC}$$

Time Constant

$$\tau = RC = 1500 \times 100 \text{ pF} = 150 \text{ ns}$$

This tells us how fast the discharge happens.

Peak Current

$$I_{\text{peak}} = \frac{V_0}{R} = \frac{2000}{1500} \approx 1.33 \text{ A}$$

That's a lot of current for tiny transistors!

Energy Stored

$$E = \frac{1}{2} C V_0^2 = 0.5 \times 100 \text{ pF} \times (2000)^2 \approx 0.2 \text{ mJ}$$

Small energy, but enough to melt silicon.

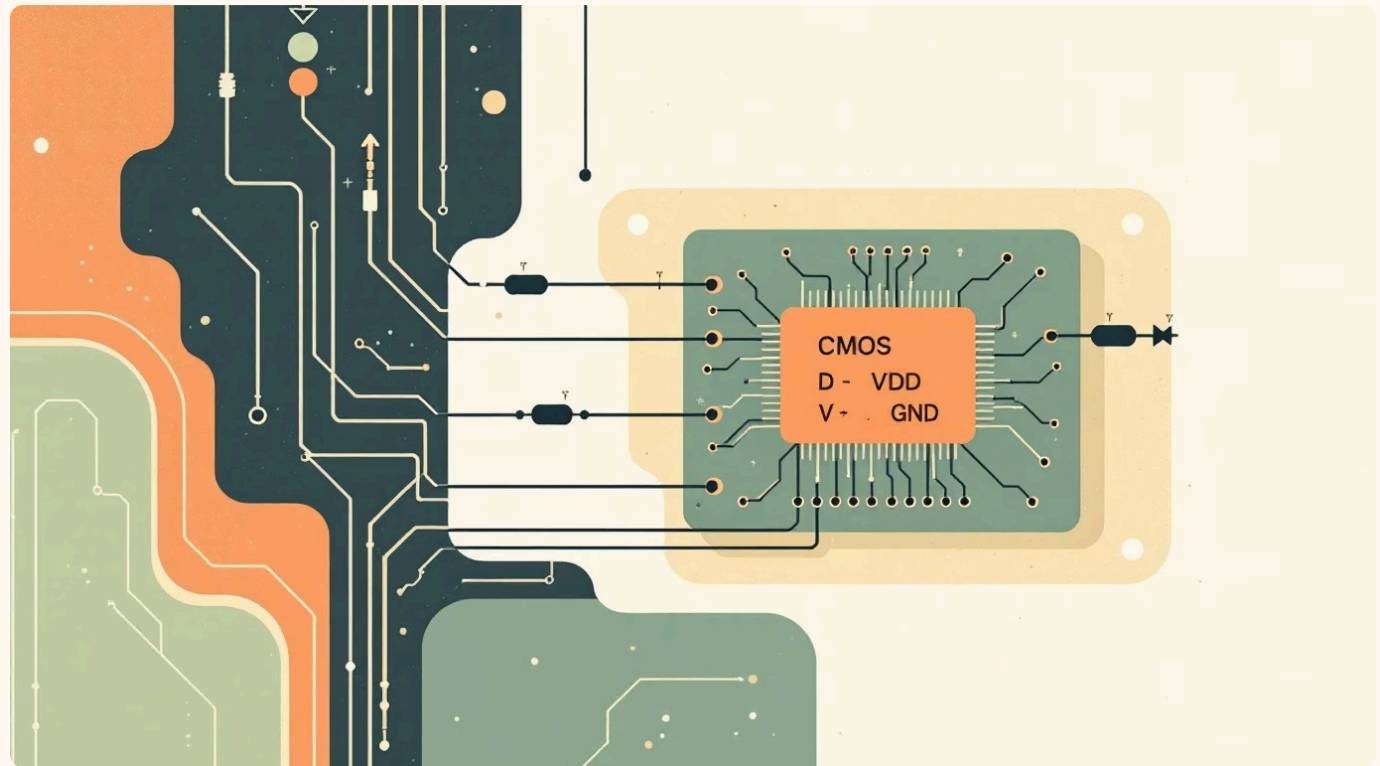
📌 **Final Results:** The current waveform is $i(t) = 1.33e^{-t/150 \text{ ns}}$ A, with peak current of ~1.33 A and total energy of ~0.2 mJ.

Protection: Double Diode Clamp

How It Works

Two diodes are connected—one to VDD and one to GND. During normal operation, they're reverse-biased and invisible to the circuit.

During a positive ESD event, the input voltage gets clamped to roughly $V_{DD} + 0.7\text{ V}$ (when the diode turns on). For negative ESD, it clamps to -0.7 V .



1

Diverts Current

Creates a low-resistance path to shunt ESD current away from sensitive transistors

2

Clamps Voltage

Limits voltage across internal circuit to safe levels ($\pm 0.7\text{ V}$ beyond supply rails)

3

Prevents Latch-Up

Fast transient + proper layout (guard rings, well contacts) stops parasitic SCR from conducting

Key Takeaways

High Current Events

HBM ESD produces ~1.33 A peak current—way beyond what CMOS transistors can handle without protection.

Energy Matters

Even 0.2 mJ can damage circuits by melting metal interconnects or gate oxides.

Diode Clamps Work

Double diode structures safely divert current and clamp voltage, protecting sensitive nodes.

Design is Crucial

Proper layout with guard rings and well contacts prevents latch-up from parasitic structures.

Contributors

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